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NPS/003/011 – TECHNICAL SPECIFICATION FOR 11 & 20kV DISTRIBUTION TRANSFORMERS.

1. Purpose

This specification details the technical requirements for Distribution Transformers for use in 11kV and 20kV ground mounted substations, or on overhead lines, that form part of the CE Electric UK electrical distribution networks.

This document supersedes the following documents, all copies of which shall be destroyed.

Ref	Version	Title
NPS/003/011	2	Technical Specification for 11 & 20kV Distribution Transformers
NETD12	1	Pole Mounted Transformers for G78 Power Supplies Operating at 11kV and 24kV and Rated Between 50kVA and 200kVA

2. Scope

This specification covers the technical requirements for Distribution Transformers for use in 11kV and 20kV ground mounted substations, or on overhead lines, that form part of the CE Electric UK distribution networks.

This specification does not apply to pad mount transformers, or to dual ratio transformers.

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3. Technical Requirements

3.1 Overview

The requirement is for Distribution Transformers that can be used on the 11kV and 20 kV networks of CE Electric UK.

Transformers are required for use in different circumstances, including:

Pole mounted – utilising single or H-Pole arrangements. HV and LV bushings may be lid mounted.

Ground mounted UK Unit Substation (UDE) style – where HV and LV connections are on the same side face of the transformer. HV and/or LV switchgear can be directly connected to the transformer, if required.

Ground mounted UK Cable Connected style – where HV and LV connections are on opposite side faces of the transformer. HV and/or LV switchgear can be directly connected to the transformer, if required.

Ground mounted units where the HV connection is made via separable cable connectors.

Regardless of configuration; arrangements of ground mounted transformers shall not have unscreened HV primary conductors.

Some transformers are required for use in particularly onerous situations including where the transformer is installed in particularly exposed conditions, or is installed near an electricity transmission tower where higher levels of insulation are required in line with Energy Networks Association Engineering Recommendation G78-2 'Recommendations for Low Voltage Supplies to Mobile Phone Base Stations with Antennae on High Voltage Structures'.

3.2 Technical Specification

3.2.1 General

The equipment shall comply with Issue 5 of the Energy Networks Association Technical Specification ENATS 35-1 Distribution Transformers (from 16kVA to 2000kVA), unless varied by this specification; in which case this specification shall take precedence.

The equipment shall also comply with the current versions of all other relevant, IEC International Standards, British Standard Specifications or equivalent Euro-Norms, and Energy Networks Association Technical Specifications (ENATS), except where varied by this standard.

3.2.2 Variations and Clarifications to ENATS 35-1

The following variations, additions or clarifications to ENATS 35-1 are referenced to the clause numbers used in ENATS 35-1 Issue 5:

3.13.1 (Definitions) Types of Transformer

Cable connected transformer - A transformer having HV and LV flanges on opposite sides of the transformer for mounting of cable boxes on the flanges. The transformer shall also be designed and constructed such that an LV fuse cabinet and/or an HV ring main unit can be directly mounted on the

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respective LV and HV flanges and assembled in such a manner that the whole substation can be handled, transported and installed as a unit.

5.6 Losses

Lifetime costs shall be calculated, for every design variant, using the formula below and the latest CE Electric UK capitalisation figures. The CE Electric UK loss £/kW figures incorporate utilisation factor and time span

$$\text{Lifetime Cost} = \text{Purchase price} + (\text{No load loss kW} \times \text{No load £/kW}) + (\text{Load loss kW} \times \text{Load loss £/kW})$$

The tenderer shall supply details of each element of this calculation, in addition to the answer.
The tenderer shall declare the maximum guaranteed loss figures for each design variant.

10.1 (Tests) General –Table 2A Insulation Levels for Pole Mounted Transformers

The test voltages for particularly exposed conditions shall generally not apply. However, transformers might be required for sites in particularly exposed conditions and the option for transformers with these enhanced insulation levels shall be available.

10.1 (Tests) General –Table 2B Insulation Levels for all other transformers

The rated Lightning Impulse (LI) for 12kV equipment shall always be at least 95kV (peak).

10.1.2 – Type Tests

The full range of type tests required by ENATS 35-1 shall be performed on, at least, the first unit of a given type and rating from a production facility.

10.1.3 – Special Tests

The following tests shall be performed on, at least, the first unit of a given type and rating from a production facility: Short-circuit withstand test in accordance with clause 10.1.3 of ENATS 35-1.

12.2 Cooling Medium

Where Distribution Transformers contain insulating fluid, this fluid shall comply fully with the current version of CE Electric UK Specification NPS/003/019 – Specification for Electrical Insulating Fluids for use in Plant & Equipment.

13.1 Surface Finish

The surface finish colour shall preferably be dark grey, or similar.

13.3 Cooling

The ability to manually position the transformer by pinch bar and rollers is not applicable to pole mounted transformers.

13.4.6 Combined drain and sampling valve

For ground mounted transformers that are not hermetically sealed; a combined drain and sampling valve shall be supplied.

13.4.8 Manual handling

The transformer design shall take into account, and allow for, methods of manual handling including, but not limited to, the use of a pinch bar and 50-75mm diameter rollers.

Ground mounted transformers shall be designed and constructed to allow the fitting of axles/wheels; and the subsequent removal, or locking, of the wheels once the transformer is in position.

Pole mounted transformers shall be designed and constructed to allow them to be lifted into position using soft slings attached to the lifting eyes. Care shall be taken in the positioning of the lifting eyes to ensure that

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the slings do not interfere or potentially cause damage to the insulator bushings when used for this purpose.

ENATS 35-1 Part 2 Cable Connected Transformers

15.1 General

The arrangement and limiting dimensions shall be as specified in Option B of Figure 5, but shall allow the close coupling of an extensible ring main unit, as illustrated in Appendix 1 below..

The design of the transformer shall allow for the HV and/or LV switchgear to be directly connected to the transformer, if required. This shall include provision for the HV flange and transformer face to accept and support either an HV cable box or a directly coupled RMU in the same manner as the Unit Transformer specified in ENATS 41-36.

The transformer shall return to the upright position when tilted 20° from the vertical, either backwards or forwards, with or without the overturning loads from the ring main unit and the LV fuse cabinet (assuming the loadings shown in fig 9 of ENATS 35-1, although with HV and LV switchgear on opposite sides of the transformer).

The transformer tank with its cover removed shall be capable of withstanding the combined load of the ring main equipment and LV distribution cabinet without distortion. The transformer will not be moved or lifted while in this condition.

Lifting fittings shall be provided, to suit slings or shackles, of adequate design to facilitate lifting in a reasonably upright position:

- (i) With or without HV and LV cable boxes.
- (ii) Complete with HV ring main equipment and/or LV distribution fuse-cabinet having weight distributions as shown in fig. 9 (but omitting the weight of the operator).

Any appropriate pair of lifting fittings shall be capable of supporting, with complete safety, the combined weight of the transformer, HV ring main equipment and LV distribution fusecabinet, when oil-filled as for service.

15.2 HV terminations

The option shall be provided for a stable fixing for HV ring main equipment complying with ENA TS 41-36 this shall be provided by supports fixed to the tank side in accordance with fig. 8.

The steelwork shall be capable of supporting the HV ring main equipment and the operator, having a load distribution as shown in fig. 9, detail 'A'. A 5mm gasket shall be provided for the transformer/ring main equipment interface. The terminal flanges shall be positioned as detailed in fig. 7. Unless otherwise stated the HV flange shall be provided with a non-returnable blanking plate of minimum thickness 5mm complete with gasket for transport purposes, together with a gasket supplied loose for permanent use.

The HV flange and supporting steelwork for the ring main unit shall be in the same plane (reference plane) in order to facilitate connection of the ring main unit. The reference plane shall be not less than 50 mm and not more than 75 mm from the tank wall.

In addition to the flange requirements of figure 8 of ENATS 41-36; the option for an HV cable box, in accordance with ENATS 12-11, shall be provided.

Where HV cable connections are made utilising separable connectors: these connectors shall be 'C' profile, DIN type 200A (non load break) separable connectors, tested in accordance with BS7888.

These connections shall be enclosed by a substantial metal cover that provides mechanical protection for the connectors and complies with clause 13.1 of this specification. This cover shall be well ventilated, but shall provide protection to class IP3XB of IEC 60529.

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15.3 LV terminations

The cast resin bushing assembly shown in BS 2562, 1979, figure 16, shall be provided. The transformer shall be capable of supporting an LV distribution fuse-cabinet complying with ENA TS 37-2 having a load distribution as shown in fig. 9, Detail 'B'. The LV cable box flange shall be positioned at the height detailed in fig. 7.

The temperature rise test shall be carried out with the transformer fitted with assemblies that represent an HV ring main unit and an LV fuse cabinet, both of which shall be close coupled to the transformer.

The option for providing an LV cable box, in accordance with ENATS 12-11, is required.

ENATS 35-1 Part 3 Unit Substation Transformers

16.2 HV terminations

The non-returnable blanking plate shall be at least 5mm thick.

In addition, the option for an HV cable box, in accordance with ENATS 12-11, shall be provided. Where HV cable connections are made utilising separable connectors: these connectors shall be 'C' profile, DIN type 200A (non load break) separable connectors, tested in accordance with BS7888. These connections shall be enclosed by a substantial metal cover that provides mechanical protection for the connectors and complies with clause 13.1 of this specification. This cover shall be well ventilated, but shall provide protection to class IP3XB of IEC 60529.

16.3 LV terminations

The cast resin bushing assembly shown in BS 2562, 1979, figure 16, shall be provided. The option for an LV cable box, in accordance with ENATS 12-11, shall be provided.

ENATS 35-1 Part 4 Pole-mounted Transformers

18.1 General

Transformer dimensions and weights shall be minimised as far as reasonably practicable. The weight limits set out in Figure 13 shall apply to platform mounted transformers. All pole-mounted transformers less than 400kg may be mounted with a single-bolt fixing arrangement.

18.2.1 HV Terminations

HV bushings shall always be supplied with arcing horns fitted.

18.2.2 LV Terminations

The LV neutral shall NOT be equipped with a surge arrestor.

18.4 Tests for pole mounted transformers

The impulse test, that is optional in clause 18.4 of ENATS 35-1 shall be performed on, at least, the first unit of a given design and rating from a production facility.

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3.3 Special Requirements

3.3.1 Pole Mounted Transformers - HV & LV Bushing Arrangements:

In addition to the HV and LV arrangements detailed in clause 18.2 of ENATS 35-1 Part 4; arrangements where both HV and LV terminations are mounted on the tank cover will be considered.

Where both HV and LV terminations are mounted on the tank cover then the design shall inherently prevent voltage transfer from HV to LV, taking account of possible causes including: vandalism, debris, birds or vermin. If this cannot be achieved practicably then the LV terminals shall be equipped with approved, individual, durable, long term UV stable, HV rated, insulating shrouds that do not compromise the performance of the LV bushings, but do provide protection against HV to LV voltage transfer. These bushing arrangements and associated features shall not compromise any performance criteria of the unit and shall not compromise the physical performance, or the ergonomic considerations, of the HV and LV connections.

Where both terminations are mounted on the tank cover then the design shall take particular account of insulation compromise as result of snowfall and the resulting accumulation of snow on and around the bushings and tank cover.

3.3.2 Ground Mounted Transformers - HV & LV Bushing Arrangements:

In addition to the Unit type and Cable Connected types detailed in ENATS 35-1; arrangements where HV and/or LV terminations are mounted on the tank cover will be considered.

Regardless of configuration; arrangements of ground mounted transformers with unscreened HV primary conductors, or with exposed LV primary conductors, are not acceptable.

3.3.3 G78 Transformers

CE Electric UK requires a small number of transformers that will be used to provide Low Voltage supplies to mobile phone base stations with antennae on High Voltage structures and these transformers shall comply with the requirements of Energy Networks Association Engineering Recommendation G78-2.

G78 installations in CE Electric UK are designed to withstand a 29kV 50Hz rise of earth potential (ROEP).

G78 transformers shall be supplied equipped with surge arrestors between the HV connections and the transformer tank. The surge arrestors shall provide the following characteristics:-

SURGE ARRESTOR CHARACTERISTICS	For use on 11kV network	For use on 20kV network
Rated Voltage	36kV	41kV
Line Discharge	Class 2	Class 2
Nominal Discharge Current	10kA	10kA
Minimum Transient Overvoltage	36kV	41.5kV
Maximum Residual Voltage (when subjected to a 10kA 8/20µs lightning current impulse)	100kV	140kV

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G78 transformers shall be supplied equipped with an external electrical bond between the neutral terminal and the transformer tank earth connection. This bond shall have a minimum cross section of 70mm² copper, or equivalent.

For G78 transformers some of the electrical characteristics specified in ENATS 35-1 clause 10.1 table 2B shall be enhanced to the following minimum insulation levels:

	LV (<1kV)	12kV	24kV
Rated lightning impulse withstand voltage (BIL, peak)	---	125kV	175kV
Rated 1 min power frequency wet withstand	3kV	36kV	50kV

3.3.4 6.6kV and 3.3kV Transformers

CE Electric UK requires a small number of transformers for use on 6.6kV and 3.3kV networks.

7.2kV transformers for use on CE Electric UK 6.6kV networks shall comply with the requirements of this NPS specification and with ENA TS 35-1

3.6 kV transformers for use on CE Electric UK 6.6kV 3.3kV networks shall have the following characteristics:

	3.6kV
Rated lightning impulse withstand voltage (LI)	40kV _{peak}
Rated power frequency withstand	10kV

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4. References

4.1. External Documentation

Reference	Title
BS 2562, 1979	Cable Boxes For Transformers And Reactors
BS 7888	LV and MV Accessories for Power Cables with Rated Voltage from 0.6/1 kV (Um = 1.2 kV) up to and Including 20.8/36 kV (Um = 42 kV)
ENA ER G78-2	Recommendations for Low Voltage Supplies to Mobile Phone Base Stations with Antennae on High Voltage Structures
ENA TS 12-11	Dry Cable Terminations In HV Switchgear For Service At Rated Voltages 12, 24 And 36kV
ENA TS 35-1 Issue 5	Distribution Transformers (from 16kVA to 2000kVA),
ENA TS 37-2	Substation Cable Distribution Boards
ENA TS 41-36	Distribution Switchgear For Service Up To 36kV (Cable And Overhead Conductor Connected)
IEC 60076	Power Transformers
IEC 60529	Degrees of Protection Provided by Enclosures

4.2. Internal documentation

Reference	Title
NPS/003/019	CE Electric UK Specification NPS/003/019 – Specification for Electrical Insulating Fluid for use in YEDL & NEDL Plant & Equipment.

4.3. Amendments from Previous Version

Clause	Subject	Amendments
2.1	Contents	Contents page added
5.6	Losses	Requirement added for manufacturer to declare maximum loss figure for each design.
3.1	Special transformers	Requirement added for some transformers to have enhanced insulation levels and surge arrestors, in line with ENA ER G78-2.
3.2.1	ENATS 35-1	Amended to reflect that the status of ENATS 35-1 has changed from draft to published and some of the clause references in ENATS 35-1 changed between draft and published version
3.2.2 (5.6)	Losses	Requirement added for loss calculations, which were previously detailed in Appendix 1, and a requirement added to declare the guaranteed maximum losses.

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Clause	Subject	Amendments
3.2.2 (10.1)	Pole Transformers	Requirement added for option of enhanced insulation levels for pole transformers for use in particularly exposed conditions.
3.2.2 (15.1)	Cable connected transformers	Requirement added for radiator arrangement on ENATS 35-1 cable connected transformers to allow use to of close-coupled extensible HV Ring Main Units and close-coupled 7-8 way LV fusecabinets.
3.2.2 (15.2)	HV terminations	Clarified requirement for 'C' profile bushings for separable connectors
3.2.2 (16.2)	HV terminations	Clarified requirement for 'C' profile bushings for separable connectors
3.3	Special requirements	New section added.
3.3.1 & 3.3.2	Cover mounted bushings	These include the option of lid mounted HV& LV bushings which were previously detailed in Appendix 1.
3.3.3	G78 transformers	This new section includes the specification for transformers in accordance with ENA ER G78-2
3.3.4	6.6kV & 3.3kV transformers	This new section includes the specification for 6.6kV and 3.3kV transformers
4.3	Amendments	New section added.

5. Definitions

Term	Definition
NONE	

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6. Authority for issue

6.1. Author

I sign to confirm that I have completed and checked this document and I am satisfied with its content and submit it for approval and authorisation.

		Sign	Date
D Blackledge	Plant & Switchgear Standards Manager	D Blackledge	06/05/2010

6.2. Policy Sponsor

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
C Holdsworth	Standards and Assessment Manager	C Holdsworth	06/05/2010

6.3. Technical Assurance

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
G Hammel	Overhead Standards Manager	G Hammel	05/05/2010

6.4. CDS Assurance

I sign to confirm that this document has been assured for issue on to the CDS system

		Sign	Date
S Johnson	CDS Administrator	S Johnson	24/05/2010

6.5. Approval

Approval is given for the content of this document

		Sign	Date
C Holdsworth	Standards and Assessment Manager	C Holdsworth	06/05/2010

6.6. Authorisation

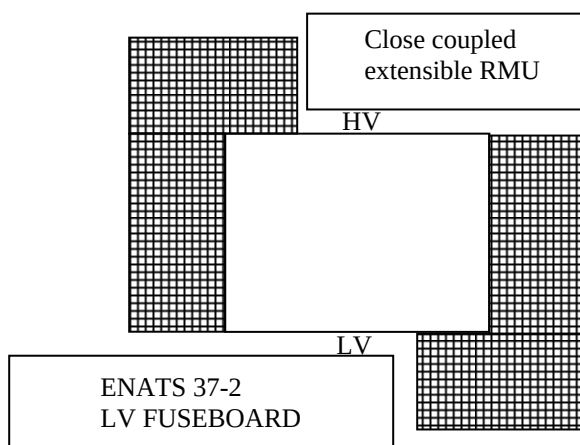
Authorisation is granted for publication of this document

		Sign	Date
M Nicholson	Head of System Strategy	M Nicholson	14/05/2010

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Appendix 1 - ENATS 'Cable Connected' Transformer Limiting Dimensions

CE Electric UK version of Option B of Figure 5 of ENATS 35-1:



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Appendix 2 - Declaration of Compliance with NPS/003/011.

NPS-003-011 Clause		Full Compliance Y or N	Details/Remarks/Description
3.2.1	ENATS 35-1 Conformance		Include completed table of self declaration of 'conformity with ENATS 35-1 Issue 5
3.2.2 – 3.13.1	'Cable Connected' transformer supports the mounting and connection of HV and/or LV switchgear		Describe fully in 14.1 below
3.2.2 – 5.6	Losses		Provide details of calculation of capitalised losses and declare guaranteed maximum losses for each variant
3.2.2 – 10.1	Option required for Table 2A higher insulation levels Table 2B Lightning Impulse Level: all 12kV units to be rated & tested 95kV		
3.2.2 – 10.1.2	Type Tests: 1 st unit – Full range of type tests		
3.2.2 – 10.1.3	Special Tests: Short circuit withstand		
3.2.2 – 12.2	Cooling medium to NPS/003/019		
3.2.2 – 13.1	Surface finish & colour		Describe fully
3.2.2 – 13.4.6	Combined drain & sampling valve		
3.2.2 – 13.4.8	Manual handling consideration Option for fitting of wheels and/or axles		Describe fully

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NPS-003-011 Clause		Full Compliance Y or N	Details/Remarks/Description
3.2.2 – 15.1	Dimensions B of Fig 5 Direct connection of HV and/or LV switchgear in similar manner to Unit Tx Return to upright from 20° Lifting/manual handling with/without HV and/or LV switchgear		Describe fully Describe fully
3.2.2 – 15.2	Direct connection of HV switchgear 5mm blanking plate & 2 off 5mm gasket Options for HV cable box		Describe fully
3.2.2 – 15.3	Direct connection of LV switchgear Cast resin bushing assembly to BS 2562:1979 fig 16 Options for LV cable box		Describe fully
3.2.2 – 16.2	5mm blanking plate & 2 off 5mm gasket Options for HV cable box		Describe fully
3.2.2 – 16.3	Cast resin bushing assembly to BS 2562:1979 fig 16 Capable of supporting LV cabinet to ENATS 37-2 Options for LV cable box		Describe fully Describe fully
3.2.2 – 18.2.1	Arcing horns required		
3.2.2 – 18.4	Impulse type tests		

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NPS-003-011 Clause		Full Compliance Y or N	Details/Remarks/Description
3.3.1	Pole transformers with lid mounted HV & LV bushings		Describe how voltage transfer between HV and LV bushings is prevented Describe how the effects of snow are mitigated.
3.3.2	Ground mounted transformers with lid mounted HV & LV bushings – no exposed primary conductors		Describe fully
3.3.3	G78 transformers: Surge arrestors required Neutral/tank bonding Enhanced BIL Enhanced power frequency withstand		

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Appendix 3 - Declaration of Compliance with IEC 60076 and ENATS 35-1 Issue 5.

CLAUSE BY CLAUSE CONFORMANCE WITH ENA TS 35-1

Transformers covered by ENA TS 35-1 shall comply with the latest issues of the relevant International and British Standards. ENA TS 35-1 is intended to amplify and/or clarify the requirements of those standards.

This check sheet identifies the clauses in ENA TS 35-1 and the clauses of IEC 60076-1. The manufacturer shall declare conformance or otherwise, clause by clause, using the following levels of conformance declaration codes.

Conformance declaration codes N/A = Clause is not applicable / appropriate to the product Cs1 = The product conforms fully with the requirements of this clause Cs2 = The product conforms partially with the requirements of this clause Cs3 = The product does not conform to the requirements of this clause	Instructions for completion - When Cs2 or Cs3 is entered, the reason for non-conformance shall be entered - If options are specified, the remark should identify the option offered by the manufacturer - The manufacturer should complete the schedule items in the Remarks column. Additional sheets may be referred to if more detail is required.
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Clause / Sub-clause	Requirement	Conformance Code IEC 60076-1	Conformance Code ENA TS 35-1	Schedule item	Remarks
1	Scope and service conditions			N/A	
1.1	Scope				
1.2	Normal Service conditions		N/A	N/A	
2	Normative references			N/A	
3	Definitions		N/A	N/A	
3.1	General				
3.2	Terminals and Neutral Point		N/A	N/A	
3.3	Windings		N/A	N/A	
3.4	Rating		N/A	N/A	

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Clause / Sub-clause	Requirement	Conformance Code IEC 60076-1	Conformance Code ENA TS 35-1	Schedule item	Remarks
3.5	Tappings		N/A	N/A	
3.6	Losses and No-load Current		N/A	N/A	
3.7	Short-circuit Impedance and Voltage Drop		N/A	N/A	
3.8	Temperature Rise		N/A	N/A	
3.9	Insulation		N/A	N/A	
3.10	Connections		N/A	N/A	
3.11	Kinds of Tests		N/A	N/A	
3.12	Meteorological Data with Respect to Cooling		N/A	N/A	
3.13	Types of Transformer	N/A		Type of transformer offered	
4	Rating		N/A	N/A	
4.1	Rating		N/A	N/A	
4.2	Loading cycle		N/A	N/A	
4.3	Preferred Values of Rated Power			Rated Power kVA Rated Voltage Ratio V	
4.4	Operation at higher than rated voltage and/or at disturbed frequency		N/A	N/A	
5	Requirements for transformers having a tapped winding		N/A	N/A	
5.1	Notation of tapping range				
5.2	Tapping voltage - tapping current, etc			N/A	
5.3	Tapping power – full power tappings			N/A	
5.4	Specification of tappings in enquiry and order		N/A	N/A	

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Clause / Sub-clause	Requirement	Conformance Code IEC 60076-1	Conformance Code ENA TS 35-1	Schedule item	Remarks
5.4.1	Single phase pole-mounted transformer rated 50kVA and under	N/A		Tapping range offered	
5.4.2	All other transformers	N/A		Tapping range offered	
5.5	Specification of short-circuit impedance			Guaranteed Impedance%	
5.5.1	Single-phase transformer arranged to give a 3-wire supply	N/A		N/A	
5.6	Load loss and temperature rise			Load loss W Temperature rise Oil °C Windings °C	
5.6.1	Dual ratio transformers			N/A	
6	Connection and phase displacement symbols for three-phase transformers		N/A	N/A	
6.1	Three phase transformers	N/A		Vector group symbol	
6.2	Single-phase transformers	N/A		N/A	
6.3	Dual ratio transformers	N/A		Method of voltage ratio change	
7	Rating plates		N/A	N/A	
7.1	Information to be given in all cases		N/A	N/A	
7.2	Additional information to be given when applicable			N/A	
8	Miscellaneous requirements			N/A	
8.1	Dimensioning of neutral connection				
8.2	Oil preservation system			Type of oil preservation system	

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Clause / Sub-clause	Requirement	Conformance Code IEC 60076-1	Conformance Code ENA TS 35-1	Schedule item	Remarks
8.3	Load rejection on generator transformers	N/A		N/A	
9	Tolerances		N/A	N/A	
10 10.1	Tests General requirements			Whether pole-mounted transformer is designed for enhanced dielectric test levels	
10.1.1	Routine tests			N/A	
10.1.2	Type tests			Sound Power Level dB(A)	
10.1.3	Special tests			Special tests included in offer	
10.2	Measurement of winding resistance		N/A	N/A	
10.3	Measurement of winding ratio and phase displacement		N/A	N/A	
10.4	Measurement of short-circuit impedance and load loss		N/A	N/A	
10.5	Measurement of no-load loss and current		N/A	Guaranteed No-load loss W	
10.6	Measurement of the harmonics of the no-load current		N/A	N/A	
10.7	Measurement of zero-sequence impedance(s) on 3-phase transformers		N/A	N/A	
10.8	Tests on on-load tapchangers	N/A		N/A	
11	Electromagnetic compatibility		N/A	N/A	
12	Transformer details	N/A		N/A	

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Clause / Sub-clause	Requirement	Conformance Code IEC 60076-1	Conformance Code ENA TS 35-1	Schedule item	Remarks
12.1	Number of phases and frequency	N/A		N/A	
12.2	Cooling medium	N/A		Cooling medium offered	
13	Construction details	N/A		Finish colour offered	
13.1	Cleaning and painting				
13.2	Terminals	N/A		N/A	
13.2.1	Position and marking of terminals				
13.3	Cooling	N/A		Method of oil preservation	
13.4	Other fittings	N/A		N/A	
13.4.1	Plain breathing device	N/A		N/A	
13.4.2	Oil level indication	N/A		N/A	
13.4.3	Tapping switch handle security	N/A		N/A	
13.4.4	Jacking lugs	N/A		N/A	
13.4.5	Earthing terminals	N/A		N/A	
13.4.6	Combined drain and sampling valve	N/A		N/A	
13.4.7	Lifting fittings	N/A		N/A	
13.4.8	Manual handling	N/A		Description of provision for manual handling	
13.4.9	Centre of Gravity	N/A		N/A	
14	Documentation	N/A		N/A	
14.1	Drawings	N/A		Electronic drawing format	
14.2	Assembly, operations and maintenance instructions	N/A		Electronic format for instruction manual	
15	Transformer fittings for cable connected transformers	N/A		Layout for cable-connected transformer	
15.1	General				
15.2	HV terminations	N/A		Type of terminations	

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Clause / Sub-clause	Requirement	Conformance Code IEC 60076-1	Conformance Code ENA TS 35-1	Schedule item	Remarks
15.3	LV terminations	N/A		Bushing plate assembly or cast resin bushing assembly	
15.4	Fittings	N/A		N/A	
16	Transformer fittings for unit substation transformers	N/A		Method of stabilising transformer	
16.1	General				
16.2	HV terminations	N/A		Description of HV blanking	
16.3	LV terminations	N/A		Bushing plate assembly or cast resin bushing assembly	
16.4	Fittings	N/A		N/A	
17	Tests for unit substation transformers	N/A		N/A	
18	Transformer fittings for pole-mounted transformers	N/A		Single bolt or platform fixing	
18.1	General				
18.2	Terminations	N/A		N/A	
18.2.1	HV terminations	N/A		Description of arcing horns/surge arrester provision	
18.2.2	LV terminations	N/A		Description of surge arrester provision	
18.3	Fittings	N/A		N/A	
19	Tests for pole-mounted transformers	N/A		Impulse test on LV included	